

[Menu](#)

UI Framework

▼ Relevant source files

`admin/utilities/downloadInteractionLog.php``h6styles.css``hclient/core/layout.js``hclient/framecontent/initPage.php``hclient/framecontent/initPageTheme.php``index.php``layout_default.js``redirects/resolver.php``viewers/record/viewRecord.php`

Purpose and Scope

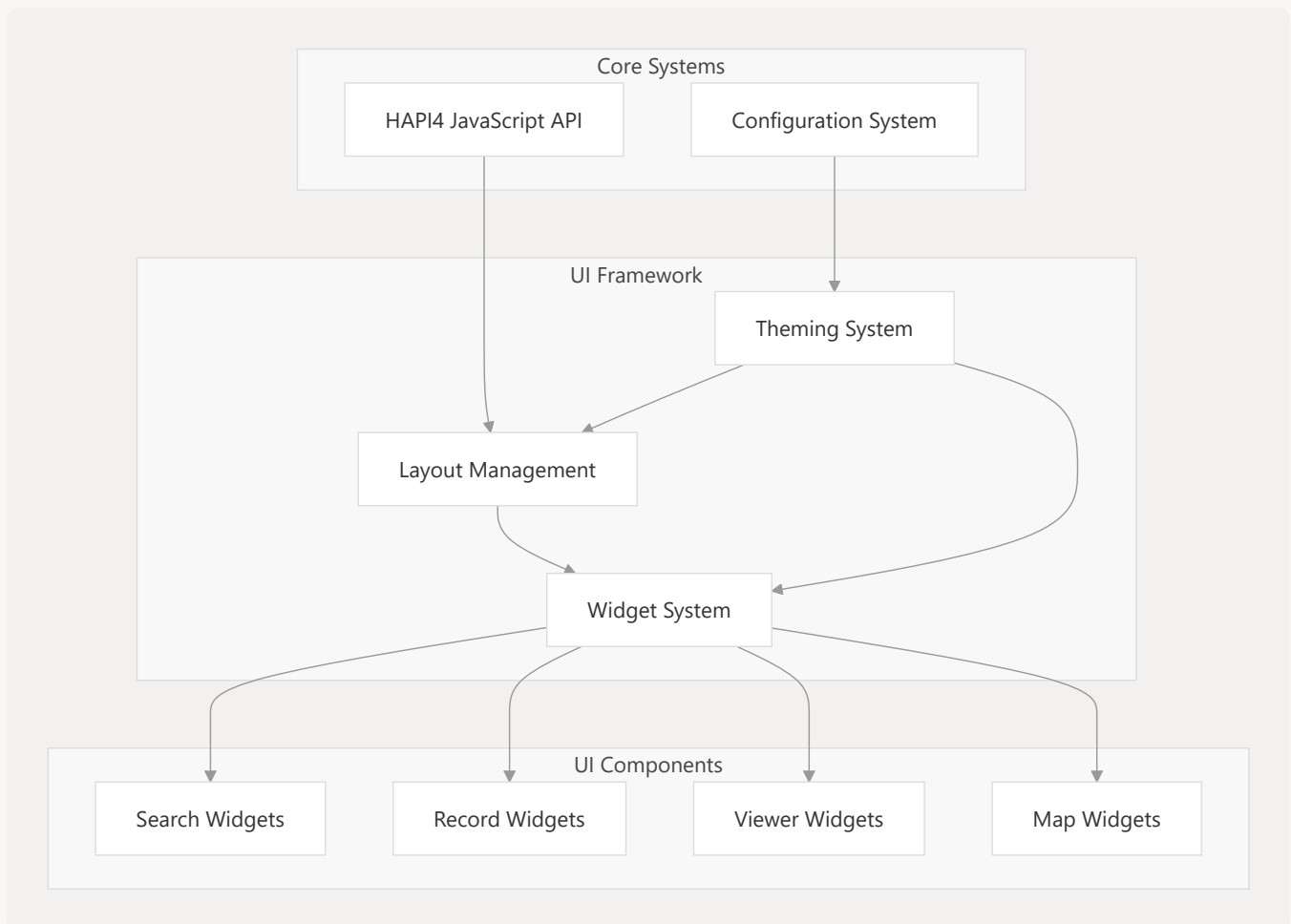
This document describes the client-side UI framework of the Heurist Network system. The framework provides a flexible, widget-based approach to building user interfaces with configurable layouts, theming, and component management. This document covers the core architecture, layout system, widget system, and theming mechanisms that together form the foundation of Heurist's user interface.

For information about specific UI components like Record Editing, see [Record Editing](#). For details on the Search Interface, see [Search Interface](#).

Architecture Overview

The Heurist UI Framework uses a widget-based architecture built on jQuery UI, with a layout management system that allows for flexible arrangement of components. The framework separates presentation concerns into three main areas:

1. **Layout Management** - Controls how UI elements are arranged on the page
2. **Widget System** - Provides reusable UI components with specific functionality
3. **Theming System** - Manages the visual appearance across the application



Sources: `index.php` 1-584 `hclient/framecontent/initPage.php` 1-476 `layout_default.js` 1-301

Layout System

The layout system manages the arrangement of widgets on the screen. It's defined in `layout_default.js` and implemented in `HLayoutMgr` and `layout.js`.

Layout Configuration

Layouts are defined in the `cfg_layouts` array in `layout_default.js`. Each layout has:

- A unique identifier
- A theme
- A layout type (free, cardinal)
- Pane definitions with widget configurations



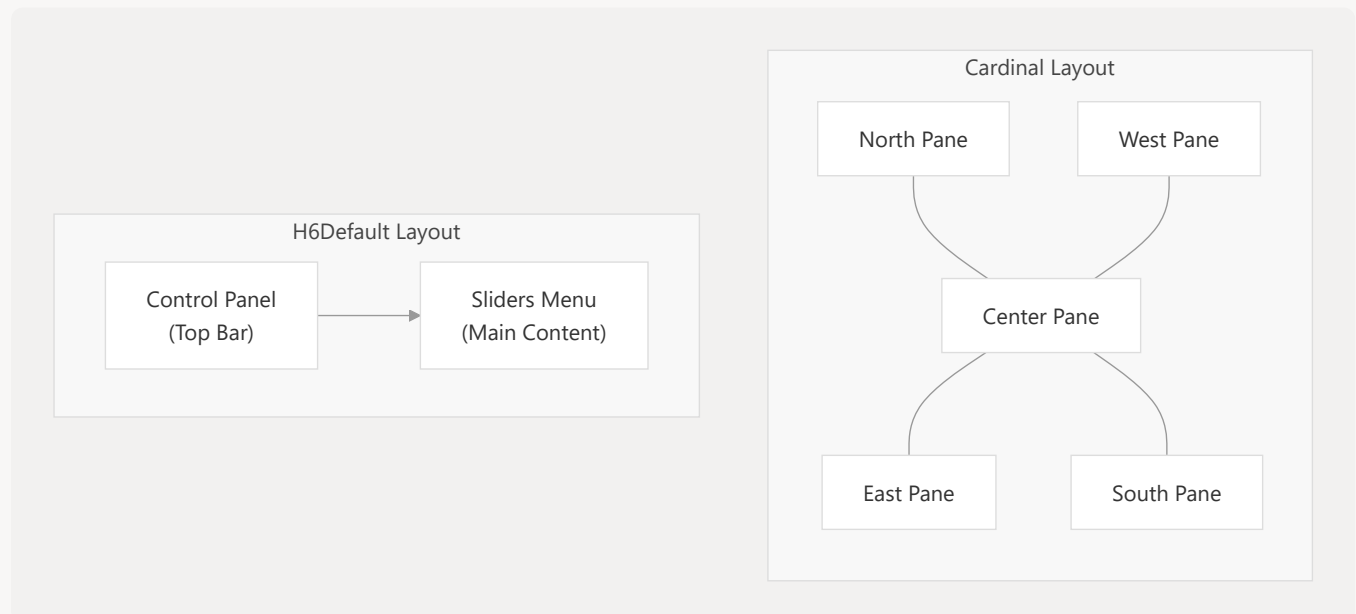
Sources: `layout_default.js` 17-301 `hclient/core/layout.js` 16-128

Layout Types

The framework supports several layout types:

1. **Cardinal Layout** - Divides the screen into North, South, East, West, and Center panes
2. **Free Layout** - Allows absolute positioning of elements
3. **Group Layout** - Organizes widgets into logical groups

The most commonly used layout is "H6Default" which combines a top control panel with a main content area organized using the slidersMenu widget.

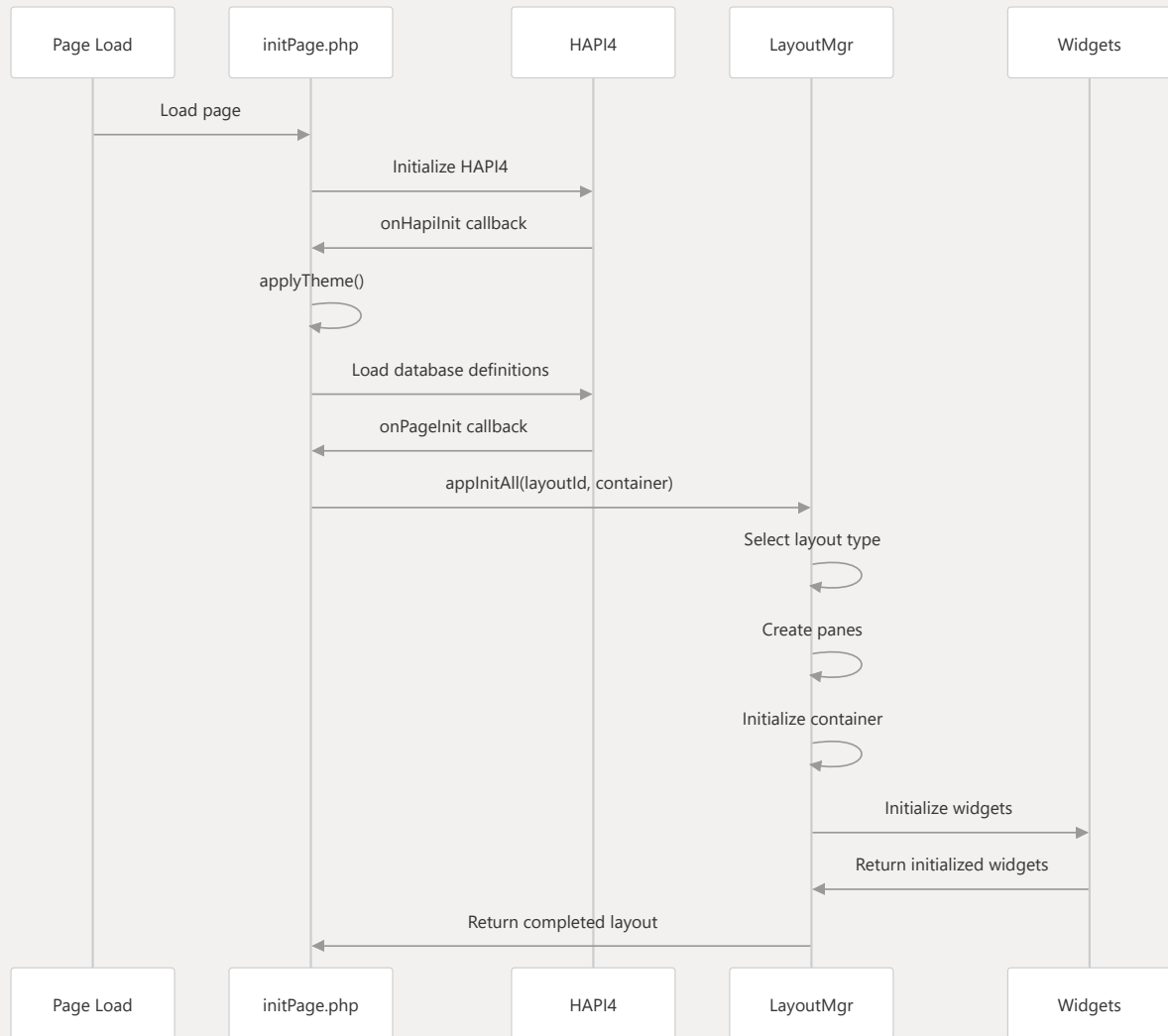


Sources: `layout_default.js` 99-247 `hclient/core/layout.js` 228-428

Layout Initialization

The layout initialization process is managed by the `HLayoutMgr` class:

1. Layout configuration is loaded from `cfg_layouts`
2. The appropriate layout is selected based on user preferences or URL parameters
3. Panes are created according to layout type
4. Widgets are initialized and added to their respective panes



Sources: `index.php` 364-420 `hclient/framecontent/initPage.php` 291-344
`hclient/core/layout.js` 219-516

Widget System

The widget system provides reusable UI components that implement specific functionality. Widgets are jQuery UI widgets that are dynamically loaded and initialized.

Widget Configuration

Widgets are defined in the `cfg_widgets` array in `layout_default.js`. Each widget configuration includes:

- A unique identifier
- A display name
- The jQuery widget name
- Path to the widget script

- Minimum size constraints

```
{id:'heurist_Search', name:'Search', widgetname:'search', script:'hclient/widgets/search/search.'}
```

Sources: `layout_default.js` 38-79 `index.php` 237-313

Widget Categories

The Heurist UI includes several categories of widgets:

1. Core UI Widgets

- Control Panel - Main navigation and user controls
- Sliders Menu - Section-based menu system

2. Search and Navigation Widgets

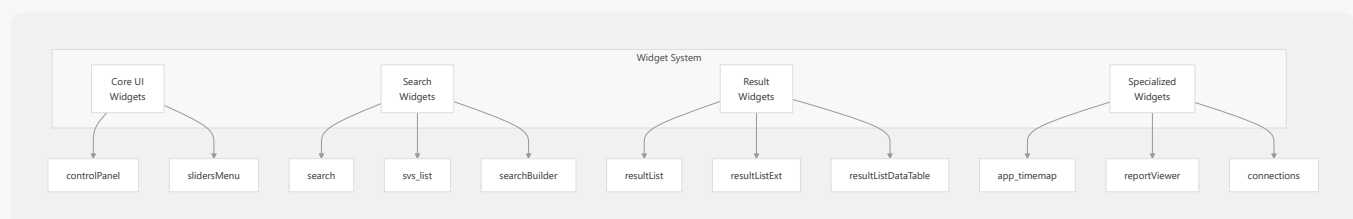
- Search - Query building and execution
- Saved Searches - Management of stored queries

3. Result Viewing Widgets

- Result List - Display search results
- Record View - Display record details

4. Specialized Widgets

- Map - Geographical visualization
- Report Viewer - Custom report templates
- Connections - Network visualization



Sources: `layout_default.js` 38-79 `index.php` 237-313

Widget Loading and Initialization

Widgets are loaded dynamically as needed:

1. The layout system identifies which widgets to load

2. Required script files are loaded
3. Widgets are initialized with their configuration options
4. Widgets are added to their respective panes

Most widgets rely on the HAPI4 JavaScript API to interact with the server and manage data.

Sources: `hclient/core/layout.js` 426-516 `index.php` 237-313

Theming System

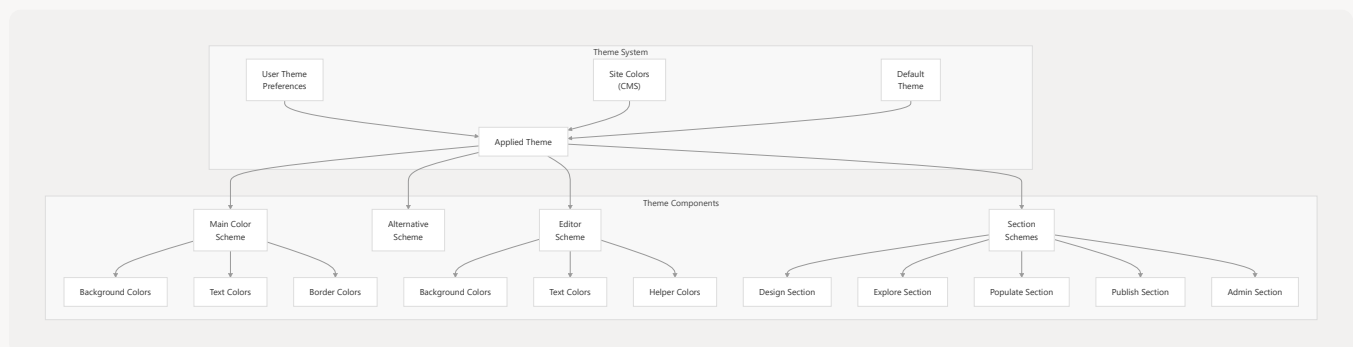
The theming system controls the visual appearance of the UI and allows for customization based on user preferences.

Theme Structure

Themes are implemented through CSS variables and rule sets, with the primary theme definition in `initPageTheme.php`. The system supports:

1. Main color scheme for general UI
2. Alternative scheme for headers
3. Editor-specific colors
4. Section-specific themes (Design, Explore, Populate, Publish, Admin)

The theming system also applies different styles to UI states (default, hover, active, pressed).



Sources: `hclient/framecontent/initPageTheme.php` 21-635 `h6styles.css` 1-303

Theme Application

Themes are applied during page initialization:

1. User preferences are loaded from the server
2. Theme settings are extracted from preferences
3. CSS variables are set based on theme settings

4. Section-specific themes are applied based on the current UI section

Version 6 of Heurist introduced a new theming approach with distinct visual styles for different functional sections (Design, Explore, Populate, Publish, Admin), each with its own color scheme.

Sources: `hclient/framecontent/initPage.php` 434-473

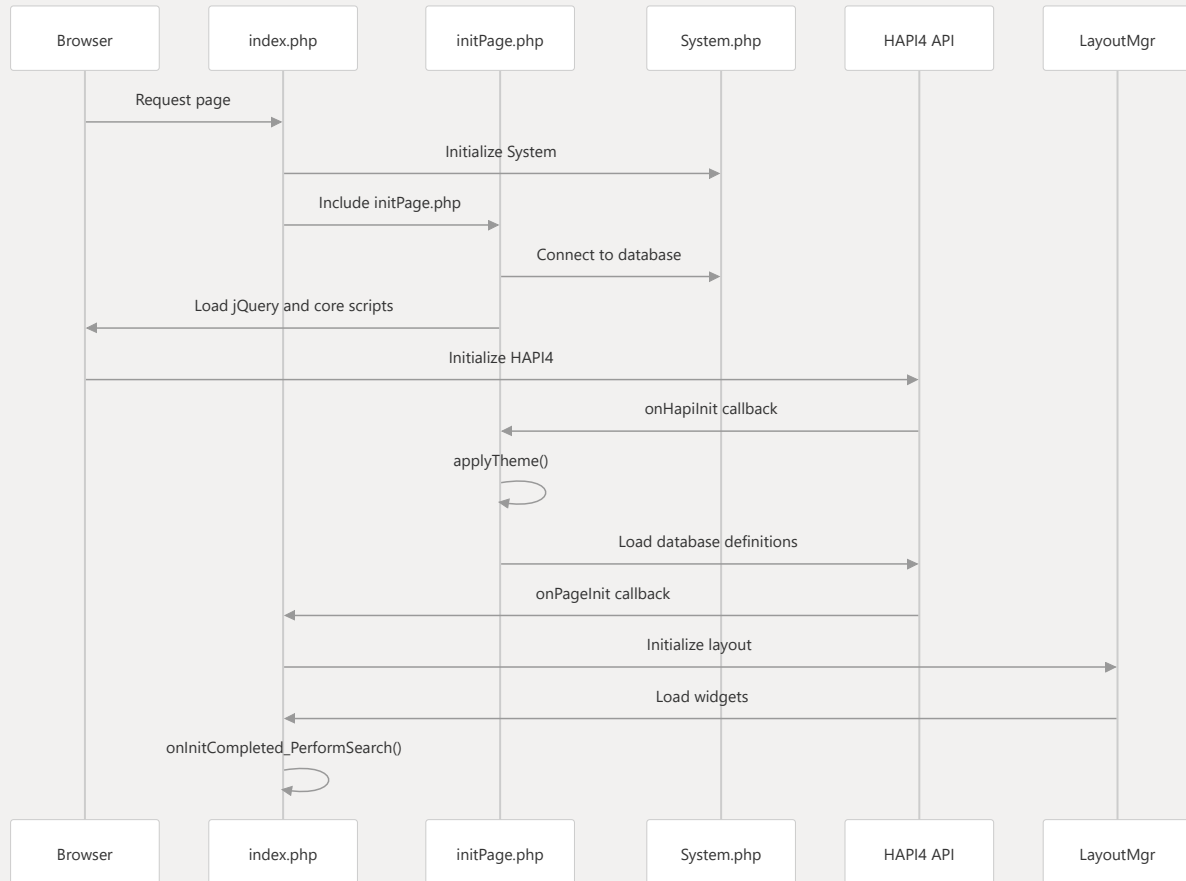
`hclient/framecontent/initPageTheme.php` 21-635

Page Initialization Process

The page initialization process ties together the layout, widget, and theming systems:

Initialization Sequence

1. The main `index.php` loads basic dependencies
2. `initPage.php` initializes the server-side system and database connection
3. Basic scripts and CSS are loaded
4. HAPI4 JavaScript API is initialized
5. Theme is applied
6. Database definitions are loaded
7. Layout is initialized with the selected configuration
8. Widgets are loaded and initialized
9. Initial search is performed (if specified)



Sources: `index.php` 364-544 `hclient/framecontent/initPage.php` 291-473

Key Functions in Initialization

- `onHapiInit()` - Called when the HAPI4 JavaScript API is initialized
- `applyTheme()` - Applies the selected theme
- `initialLoadDatabaseDefintions()` - Loads required database schema information
- `onPageInit()` - Main initialization function for the page
- `appInitAll()` - Initializes the layout and widgets

Sources: `hclient/framecontent/initPage.php` 311-435 `index.php` 366-543

UI Framework Evolution

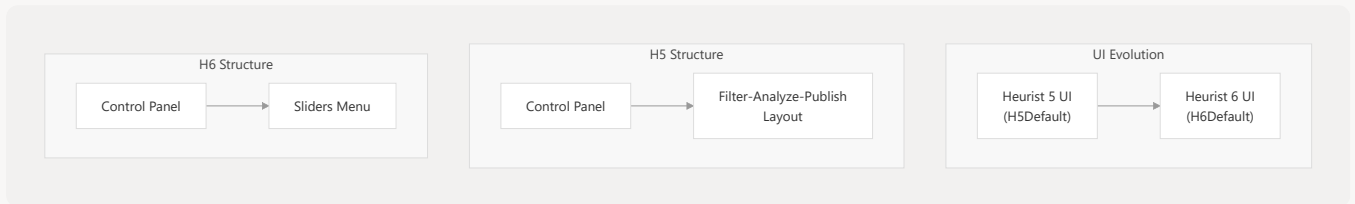
The Heurist UI has evolved through several major versions, with the current version (6) introducing significant changes:

Version 6 Improvements

1. New Sliders Menu for main navigation
2. Section-specific theming (Design, Explore, Populate, Publish, Admin)

3. Improved responsiveness and layout management
4. Enhanced widget system with better modularity

The system maintains backward compatibility while supporting the newer interface paradigms.



Sources: `layout_default.js` 103-128 `h6styles.css` 1-303

Conclusion

The Heurist UI Framework provides a flexible, configurable foundation for building complex user